

KARRI SHYAM PRASAD

Senior Data Engineer | Cloud Data Pipelines | Cloud Data Platform

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PROFESSIONAL SUMMARY

Senior GCP Data Engineer with 5+ years of production experience designing, building, and optimizing high-performance data pipelines on Google Cloud Platform. Deep expertise in BigQuery schema design, Cloud Composer/Airflow orchestration, GCS-based medallion architecture, and Dataflow-based batch and streaming ingestion. Most recent work at HCA Healthcare involved migrating mainframe and Teradata workloads to GCP, replacing legacy Zeke/DMX scheduling with 100+ monitored Airflow DAGs, and implementing Pub/Sub-triggered near-real-time event pipelines in production healthcare environments with strict HIPAA data quality and governance requirements. Skilled Python and SQL developer with hands-on CI/CD (GitHub Actions), REST API development (FastAPI), and data lineage documentation experience across GCP, AWS, and Azure.

TECHNICAL SKILLS

Programming & Query Languages: Python (advanced) · SQL / T-SQL / PL/SQL · PySpark · Spark SQL · HiveQL · Bash/Shell

Cloud Platforms: GCP: BigQuery · Cloud Composer · Dataflow · Pub/Sub · GCS · Cloud Run · DataStream · Vertex AI · AWS: S3 · Glue · EMR · Redshift · Lambda · Athena · Azure: Databricks · ADF · ADLS

Data Engineering & ETL: Apache Spark · Dataflow · SSIS · PySpark ETL Frameworks · Pub/Sub Streaming · CDC Pipelines · REST API (FastAPI) · Great Expectations · Data Reconciliation

Orchestration & Workflow: Apache Airflow (Cloud Composer) · DAG Authoring · Custom Operators · SLA Alerting · Retry Logic · Autosys · Zeke · Control-M

Databases & Data Warehousing: BigQuery · Snowflake · Teradata · Redshift · Oracle · SQL Server · Hive · Delta Lake · Apache Iceberg

Streaming Technologies: Apache Kafka · GCP Pub/Sub · Event-Driven Architecture · Near Real-Time Ingestion · Dead-Letter Handling

DevOps & CI/CD: GitHub Actions · Jenkins · Docker · Kubernetes (GKE / EKS / AKS) · Terraform · Azure DevOps · Oauth

Data Modeling & Governance: Dimensional Modeling (Kimball) · Star/Snowflake Schema · SCD Type 1/2/3 · Medallion Architecture · Data Vault 2.0 · Column-Level Encryption · Row-Level Security · Data Lineage · Metadata Management

Monitoring & Observability: Prometheus · Grafana · Airflow SLA Alerts · Pipeline Health Dashboards · Incident Management · Root Cause Analysis

Visualization & Reporting: Qlik · Power BI · Tableau · SQL Views / Semantic Layer Design · KPI Delivery

PROFESSIONAL EXPERIENCE

Senior GCP Data Engineer — Enterprise Data Platform

June 2024 – Present

HCA Healthcare Remote

- Architected and executed migration of legacy mainframe batch jobs and Teradata/Oracle on-prem workloads to GCP — rewrote transformation logic in Python and PySpark on Dataflow and deployed 100+ production Airflow DAGs in Cloud Composer with dependency chaining, SLA alerting, retry logic, and dead-letter handling.
- Designed BigQuery schemas for enterprise clinical datasets applying partitioning by ingestion date and clustering on high-cardinality filter columns — eliminating full table scans and reducing query slot consumption on critical reporting workloads.
- Built GCS-based medallion/lakehouse architecture (raw landing, cleansed, analytics-ready layers) with Parquet-formatted output; implemented Pub/Sub-triggered Cloud Run jobs supporting near-real-time event ingestion from clinical source systems with sub-minute latency.
- Developed Python ETL frameworks deployed through GitHub Actions CI/CD, replacing fragmented shell scripts and DMX definitions — introduced version control, automated testing, and structured deployment gates across 15+ previously unmanaged pipelines.
- Engineered REST APIs using FastAPI to expose curated datasets to downstream analytics and Data Science teams, including a RAG-based metadata discovery assistant (LangChain, FAISS, HuggingFace embeddings) deployed on Cloud Run.
- Ensured quality of critical clinical data elements using Great Expectations validation and automated reconciliation checks; documented end-to-end data lineage and operational metadata as part of the enterprise governance initiative.
- Implemented column-level encryption and row-level access controls in BigQuery for HIPAA-sensitive fields — aligning pipeline outputs with regulatory compliance and audit requirements.
- Integrated pipeline outputs with Qlik and Power BI reporting layers, designing the semantic SQL view layer (grain, aggregation logic, SCD handling) feeding enterprise dashboards for clinical operations and finance teams.

Data Engineer — Capital Markets & Credit Risk Analytics July 2023 – May 2024

Citi Irving, TX

- Built and maintained PySpark and Hive-based ETL pipelines on a Hadoop cluster ingesting 2M+ daily financial transactions from trading systems and Bloomberg/Reuters market data feeds, with idempotent execution ensuring accurate reruns after upstream failures.
- Led EDW-to-EAP migration, moving mortgage and banking datasets from Teradata into the Enterprise Analytics Platform — rewrote HiveQL transformation logic for the Spark stack and produced automated reconciliation reports validating row counts and aggregations between source and target.

- Authored 35+ Autosys job flows for credit risk batch workflows with conditional branching — downstream PD/LGD/EAD jobs held and triggered on-call alerts if market data feeds failed to land, preventing processing on stale data.
- Developed T-SQL stored procedures for credit risk calculations (PD, LGD, EAD) across 500K+ loan accounts using CTEs and window functions — replaced cursor-based loops with set-based logic, reducing overnight batch runtime from 6 hours to ~90 minutes.
- Applied statistical outlier detection (z-score and IQR methods) to incoming market data before load — intercepting bad ticks upstream before they reached downstream risk models and regulatory reporting figures.
- Established Jenkins CI/CD pipelines for Python and SQL deployments with automated PyTest coverage gates (80% minimum) — enforcing code quality standards across the credit risk engineering team.
- Performed root cause analysis on recurring pipeline failures, documented remediation runbooks, and collaborated with quant analysts to translate risk formulas into production Python with edge-case handling.

Cloud Platform & Data Engineer — Financial Services*August 2019 – January 2022*

Accenture *Bengaluru, India*

- Provisioned and managed Kubernetes clusters across EKS, GKE, and AKS for financial services clients — wrote Terraform modules for cluster setup, node pools, IAM roles, and network policies, creating reproducible and auditable infrastructure deployments.
- Built CI/CD pipelines in Jenkins and GitHub Actions for application and data platform deployments — containerized ETL workloads with Docker, implemented staging/prod promotion gates, and enforced branch protection rules.
- Developed Python and SSIS ETL pipelines for a core banking client processing financial transaction data — designed transformation logic, handled nulls/duplicates, and integrated Autosys scheduling for 20+ interdependent month-end financial close workflows.
- Integrated CyberArk secrets management across pipeline deployments — eliminated credential storage in config files and addressed recurring audit findings for financial services clients.
- Deployed Prometheus and Grafana monitoring stacks for cluster and pipeline health — providing data and platform teams with real-time visibility into job status, resource consumption, and SLA compliance.

KEY PROJECTS

Mainframe-to-GCP Pipeline Migration — HCA Healthcare

Led end-to-end migration of mainframe DMX batch jobs and Zeke-scheduled workflows to GCP. Reverse-engineered legacy definitions, rewrote logic in Python/PySpark, deployed as Airflow DAGs in Cloud Composer, and built GCS medallion architecture with BigQuery as the serving layer — covering clinical and operational datasets feeding Qlik/Power BI dashboards.

RAG-Based Metadata Discovery System — HCA Healthcare

Built an internal analyst-facing tool using LangChain, FAISS vector store, HuggingFace embeddings, and Groq API for natural-language metadata search across enterprise datasets. FastAPI backend deployed on Cloud Run via Docker — adopted at HCA to reduce analyst dependency on engineering for data discovery.

EDW-to-EAP Migration — Mortgage & Banking Data (Citi)

Migrated mortgage and capital markets datasets from Teradata EDW to the Enterprise Analytics Platform. Rewrote HiveQL for Hadoop/Spark stack, built automated reconciliation reports for row-count and aggregation validation, and coordinated cutover with risk reporting teams.

EDUCATION

M.S. Computer Information Systems — Florida Institute of Technology

B.E. Electronics and Communication Engineering — Gandhi Institute of Technology and Management

CERTIFICATIONS

- AWS Certified Solutions Architect – Associate (July 2023 – July 2026)
- Microsoft Certified: Azure Data Engineer Associate (March 2024)
- Snowflake SnowPro Core Certification — In Progress
- Certified Data Science Professional — DataCamp (September 2023)